Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	29 June 2026
Team ID	
Project Name	Strategic Product Placement Analysis
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Strategic Product Placement Analysis using Machine Learning and Data Analytics

Reference: https://cloud.google.com/architecture

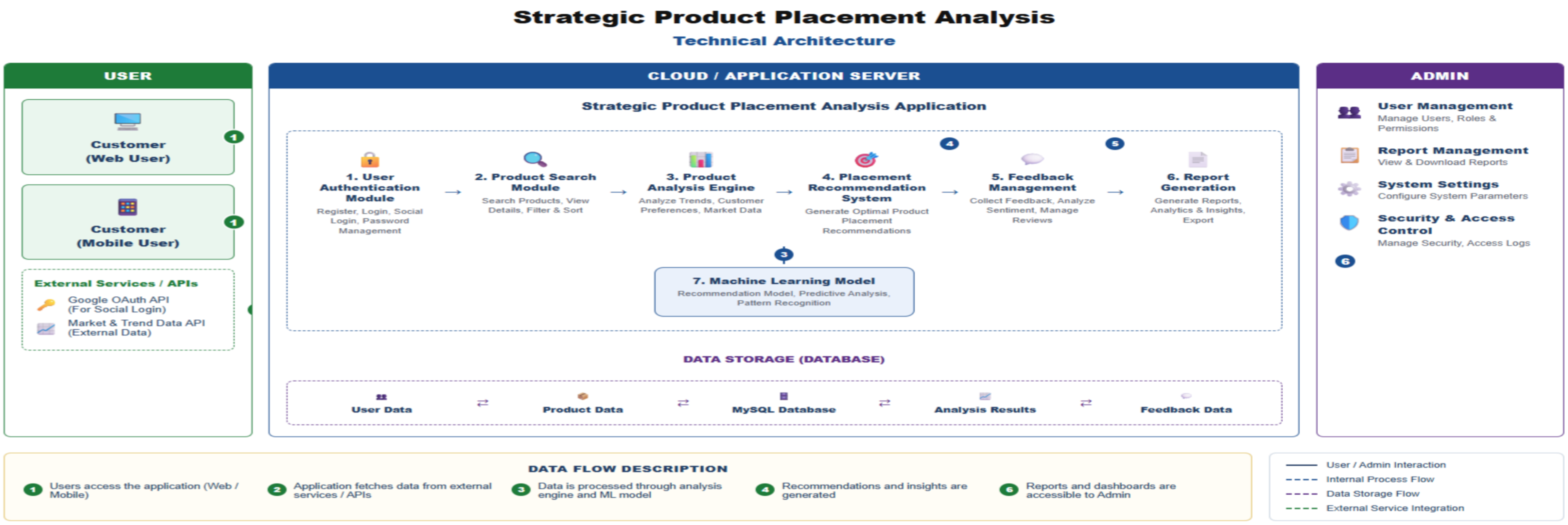

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	Data Collection Module	Collects retail transaction, product, customer, and store data from datasets.	CSV, Kaggle Dataset, Python
2	Data Preprocessing Module	Cleans missing values, removes duplicates, encodes categorical features, and prepares data for analysis.	Pandas, NumPy
3	Exploratory Data Analysis (EDA)	Analyzes sales trends, customer purchasing behavior, and product performance through visualizations.	Matplotlib, Seaborn, Plotly
4	Feature Engineering Module	Creates new features such as sales growth, product popularity, shelf category, and seasonal indicators.	Python, Pandas
5	Product Placement Analysis Engine	Identifies relationships between product location, customer behavior, and sales performance.	Python, Scikit-learn
6	Machine Learning Prediction Model	Predicts the most effective product placement strategy for maximizing sales.	Random Forest, XGBoost, Decision Tree
7	Visualization Dashboard	Displays KPIs, sales trends, heatmaps, and placement recommendations.	Power BI / Tableau / Streamlit
8	Database Management	Stores retail data, processed datasets, prediction results, and reports.	MySQL

9	Report Generation Module	Generates sales reports, product performance summaries, and strategic recommendations.	Python, ReportLab, Pandas
10	Deployment Module	Hosts the application for users to analyze retail data and view recommendations.	Flask, Streamlit, Render

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Data Processing	Cleans, transforms, and prepares retail sales data for analysis and model training.	Python, Pandas, NumPy
2	Machine Learning	Predicts optimal product placement and identifies factors influencing sales performance.	Scikit-learn, XGBoost, Random Forest
3	Data Visualization	Presents sales trends, customer behavior, and product performance through interactive charts and dashboards.	Matplotlib, Seaborn, Plotly, Power BI
4	Scalability	Efficiently processes large retail datasets and supports future data expansion.	Python, MySQL
5	Performance	Delivers fast data analysis and prediction using optimized preprocessing and machine learning algorithms.	Pandas Optimization, NumPy, Scikit-learn

References:

https://c4model.com/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d

https://flask.palletsprojects.com/

https://cloud.google.com/architecture